

ECO 530 - Poster Draft 2

Fall 2024 - Ibrahim Quansah

```
#Loading Libraries
```

```
library(tidyverse)
library(kableExtra)
library(vtable)
library(modelsummary)
library(dplyr)
library(readr)
library(modelsummary)
library(ggplot2)
library(knitr)
library(tinytex)
library(tseries)
library(estimatr)
library(car)
library(lmtest)
```

Research Question

Is there any relationship between U.S. economic growth (GDP) and trade volumes (Net Trade)?

Purpose: To identify whether a correlation exists between the two variables, meaning that changes in trade volumes might correlate with changes in economic growth (GDP).

Introduction

This study explores the relationship between economic growth, measured by Gross Domestic Product (GDP), and trade volumes, specifically net trade, in the United States. Net trade refers to the difference between a country's exports and imports, and it plays a crucial role in shaping the overall economic performance of a nation. In the context of the U.S., a key player in global trade, understanding how changes in trade volumes influence economic growth is critical for evaluating economic policies and trade agreements.

I will utilize data on U.S. GDP and Net Trade from the World Bank DataBank, covering two decades. The GDP represents the total economic output of the country, while net trade captures the balance of goods and services traded internationally.

Data

```
#Reading Net Trade File
Net_Trade <- read_csv("~/ECO 530/Data/Net Trade.csv")
head(Net_Trade)
```

```
## # A tibble: 6 × 2
##   DATE      `Net Trade`
##   <chr>      <dbl>
## 1 1/1/1993    -6777
## 2 4/1/1993    -6929
## 3 7/1/1993    -7031
## 4 10/1/1993   -5719
## 5 1/1/1994   -6466
## 6 4/1/1994   -8380
```

```
#Reading GDP File
GDP <- read_csv("~/ECO 530/Data/GDP.csv")
head(GDP)
```

```
## # A tibble: 6 × 2
##   DATE      GDP
##   <chr>    <dbl>
## 1 1/1/1993  6729.
## 2 4/1/1993  6809.
## 3 7/1/1993  6882.
## 4 10/1/1993 7014.
## 5 1/1/1994  7116.
## 6 4/1/1994  7247.
```

```
#Combining GDP and Net Trade in one data
Data <- merge(Net_Trade, GDP, by = 'DATE', all = TRUE)

Data$DATE <- as.Date(Data$DATE, format = "%m/%d/%Y")

# Sort chronologically
Data <- Data %>% arrange(DATE)

#Formatting Table with Kable
Data %>%
  kable(caption = "U.S. GDP and Net Trade (1993-2023)") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"))
```

U.S. GDP and Net Trade (1993-2023)

DATE	Net Trade	GDP
1993-01-01	-6777	6729.459
1993-04-01	-6929	6808.939
1993-07-01	-7031	6882.098
1993-10-01	-5719	7013.738
1994-01-01	-6466	7115.652

DATE	Net Trade	GDP
1994-04-01	-8380	7246.931
1994-07-01	-8310	7331.075
1994-10-01	-8456	7455.288
1995-01-01	-9708	7522.289
1995-04-01	-10623	7580.997
1995-07-01	-5924	7683.125
1995-10-01	-6271	7772.586
1996-01-01	-7756	7868.468
1996-04-01	-7460	8032.840
1996-07-01	-10195	8131.408
1996-10-01	-10475	8259.771
1997-01-01	-8383	8362.655
1997-04-01	-7154	8518.825
1997-07-01	-9269	8662.823
1997-10-01	-11289	8765.907
1998-01-01	-12880	8866.480
1998-04-01	-13247	8969.699
1998-07-01	-14820	9121.097
1998-10-01	-14745	9293.991
1999-01-01	-17719	9411.682
1999-04-01	-22892	9526.210
1999-07-01	-22699	9686.626
1999-10-01	-25939	9900.169
2000-01-01	-30557	10002.179
2000-04-01	-30405	10247.720
2000-07-01	-33333	10318.165
2000-10-01	-33420	10435.744
2001-01-01	-32812	10470.231
2001-04-01	-29151	10599.000
2001-07-01	-30217	10598.020

DATE	Net Trade	GDP
2001-10-01	-27488	10660.465
2002-01-01	-31161	10783.500
2002-04-01	-34732	10887.460
2002-07-01	-36088	10984.040
2002-10-01	-43607	11061.433
2003-01-01	-43806	11174.129
2003-04-01	-39696	11312.766
2003-07-01	-41842	11566.669
2003-10-01	-44080	11772.234
2004-01-01	-46754	11923.447
2004-04-01	-54169	12112.815
2004-07-01	-51686	12305.307
2004-10-01	-54918	12527.214
2005-01-01	-53134	12767.286
2005-04-01	-58152	12922.656
2005-07-01	-64882	13142.642
2005-10-01	-64568	13324.204
2006-01-01	-62397	13599.160
2006-04-01	-63800	13753.424
2006-07-01	-65095	13870.188
2006-10-01	-61882	14039.560
2007-01-01	-62183	14215.651
2007-04-01	-59997	14402.082
2007-07-01	-58158	14564.117
2007-10-01	-58290	14715.058
2008-01-01	-60200	14706.538
2008-04-01	-60889	14865.701
2008-07-01	-60292	14898.999
2008-10-01	-43967	14608.209
2009-01-01	-30193	14430.902

DATE	Net Trade	GDP
2009-04-01	-27152	14381.236
2009-07-01	-35247	14448.882
2009-10-01	-39426	14651.249
2010-01-01	-40271	14764.610
2010-04-01	-47316	14980.193
2010-07-01	-44333	15141.607
2010-10-01	-41721	15309.474
2011-01-01	-44312	15351.448
2011-04-01	-49882	15557.539
2011-07-01	-43477	15647.680
2011-10-01	-49473	15842.259
2012-01-01	-49749	16068.805
2012-04-01	-42941	16207.115
2012-07-01	-38794	16319.541
2012-10-01	-36334	16420.419
2013-01-01	-34314	16648.189
2013-04-01	-34426	16728.687
2013-07-01	-39917	16953.838
2013-10-01	-33636	17192.019
2014-01-01	-42940	17197.738
2014-04-01	-38895	17518.508
2014-07-01	-41672	17804.228
2014-10-01	-42838	17912.079
2015-01-01	-48732	18063.529
2015-04-01	-42425	18279.784
2015-07-01	-41641	18401.626
2015-10-01	-41003	18435.137
2016-01-01	-36801	18525.933
2016-04-01	-41719	18711.702
2016-07-01	-37004	18892.639

DATE	Net Trade	GDP
2016-10-01	-40976	19089.379
2017-01-01	-41124	19280.084
2017-04-01	-42579	19438.643
2017-07-01	-41593	19692.595
2017-10-01	-46535	20037.088
2018-01-01	-43508	20328.553
2018-04-01	-44811	20580.912
2018-07-01	-50563	20798.730
2018-10-01	-55212	20917.867
2019-01-01	-48097	21111.600
2019-04-01	-48568	21397.938
2019-07-01	-44479	21717.171
2019-10-01	-43943	21933.217
2020-01-01	-43724	21727.657
2020-04-01	-49402	19935.444
2020-07-01	-58182	21684.551
2020-10-01	-62022	22068.767
2021-01-01	-66105	22656.793
2021-04-01	-71488	23368.861
2021-07-01	-77095	23921.991
2021-10-01	-80792	24777.038
2022-01-01	-101914	25215.491
2022-04-01	-81215	25805.791
2022-07-01	-71217	26272.011
2022-10-01	-70762	26734.277
2023-01-01	-60321	27164.359
2023-04-01	-64806	27453.815
2023-07-01	-62222	27967.697
2023-10-01	-64915	28296.967

```

# Summary statistics for GDP and Net Trade

Summary_Table <- Data %>%
  summarise(
    GDP_Mean = mean(GDP),
    GDP_Median = median(GDP),
    GDP_SD = sd(GDP),
    Net_Trade_Mean = mean(`Net Trade`),
    Net_Trade_Median = median(`Net Trade`),
    Net_Trade_SD = sd(`Net Trade`)
  )

# Creating a Table
Table <- Summary_Table %>%
  pivot_longer(cols = everything(), names_to = "Metric", values_to = "Value")

# Displaying the table with kableExtra
Table %>%
  kable("html") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"))

```

Metric	Value
GDP_Mean	14909.861
GDP_Median	14586.163
GDP_SD	5566.641
Net_Trade_Mean	-40097.403
Net_Trade_Median	-41781.500
Net_Trade_SD	20027.065

Table

```

## # A tibble: 6 × 2
##   Metric      Value
##   <chr>      <dbl>
## 1 GDP_Mean    14910.
## 2 GDP_Median  14586.
## 3 GDP_SD      5567.
## 4 Net_Trade_Mean -40097.
## 5 Net_Trade_Median -41782.
## 6 Net_Trade_SD   20027.

```

Creating a line plot for GDP and Net Trade over time

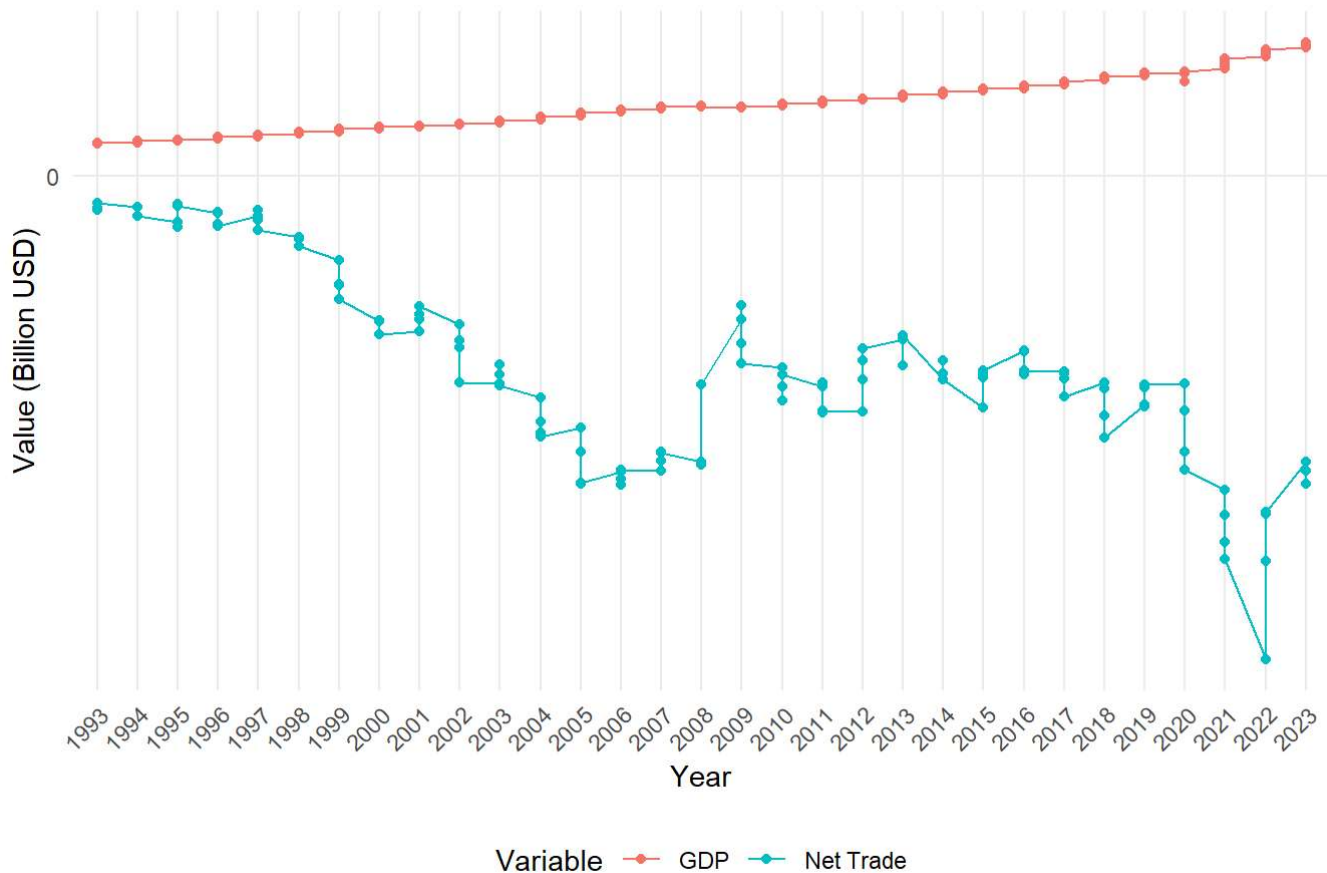
```

# Melt the data to long format for ggplot2
Data$DATE <- as.Date(Data$DATE, format = "%m/%d/%Y")
Data$Years <- format(Data$DATE, "%Y")
Data_long <- Data %>%
  pivot_longer(cols = c(GDP, `Net Trade`), names_to = "Variable", values_to = "Value") %>%
  select(Years, Variable, Value)

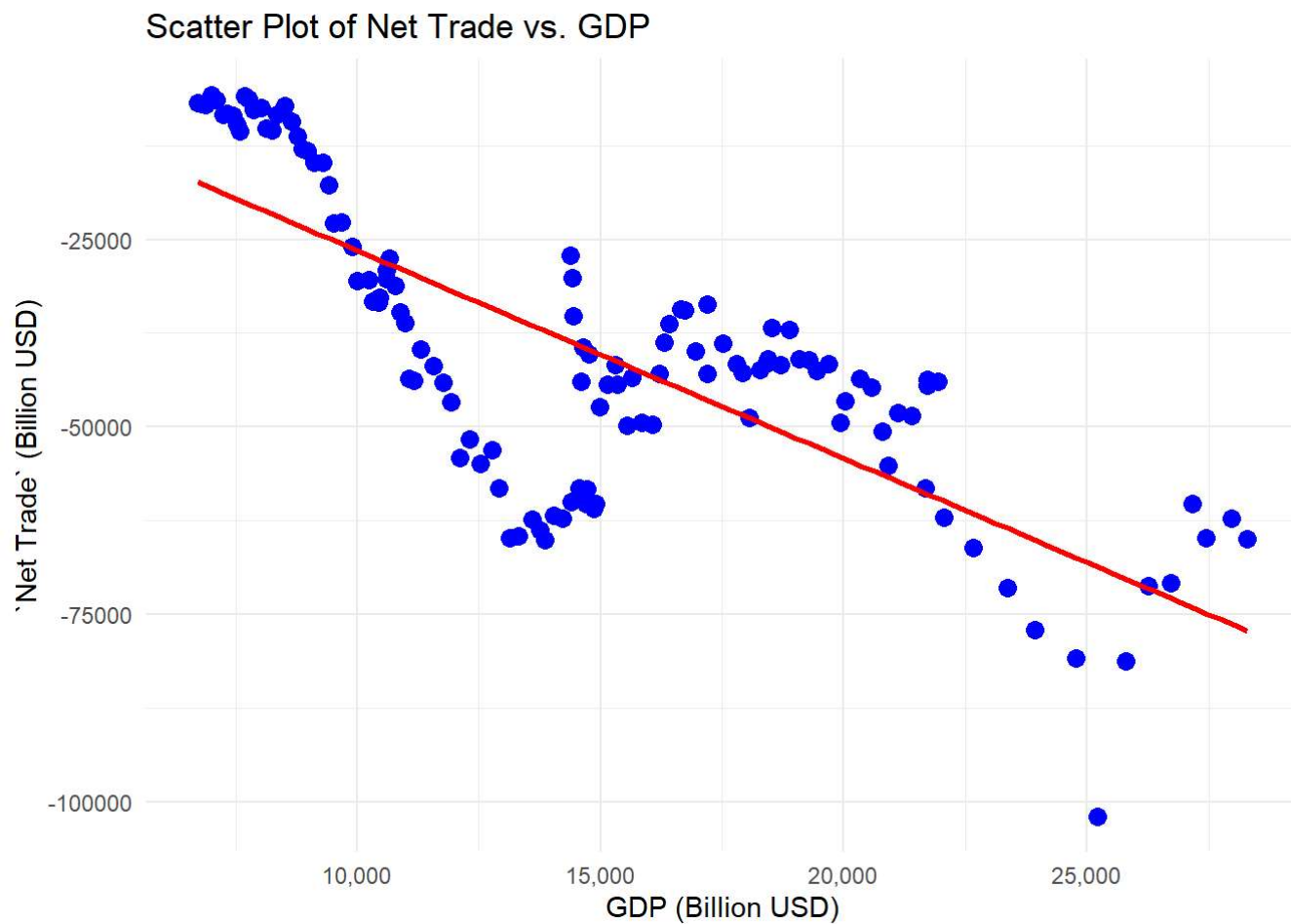
# Create the Line plot
ggplot(Data_long, aes(x = Years, y = Value, color = Variable, group = Variable)) +
  geom_line() +
  geom_point() +
  labs(title = "GDP and Net Trade Over Time", x = "Year", y = "Value (Billion USD)") +
  scale_y_continuous(labels = scales::comma, breaks = seq(0, 30000000, 5000000)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "bottom")

```

GDP and Net Trade Over Time



```
#Creating Scatter Plot of Net Trade and GDP
ggplot(Data, aes(x = GDP, y = `Net Trade`)) +
  geom_point(color = "blue", size = 3) + # Points in blue
  labs(title = "Scatter Plot of Net Trade vs. GDP",
        x = "GDP (Billion USD)",
        y = "`Net Trade` (Billion USD)") +
  theme_minimal() + # Use a minimal theme
  scale_x_continuous(labels = scales::comma)+
  geom_smooth(method = "lm", color = "red", se = FALSE)
```



```
# Calculate correlation between GDP and Net Trade
correlation <- cor(Data$GDP, Data$`Net Trade`)

# Display the correlation value
correlation
```

```
## [1] -0.7703899
```

```

#Reading GDPDEF File
GDPDEF <- read_csv("~/ECO 530/Data/GDPDEF.csv")

#Reading Interest Rate File
Interest_Rate <- read_csv("~/ECO 530/Data/FEDFUNDS.csv")

#Reading Government Expenditure File
Gov_Expen <- read_csv("~/ECO 530/Data/Gov Expen.csv")
colnames(Gov_Expen)[colnames(Gov_Expen) == "W068RCQ027SBEA"] <- "Government Expenditure"

#Reading Unemployment Rate File
UNRate <- read_csv("~/ECO 530/Data/UNRate.csv")

#Reading Gross Private Domestic Investment (GPDI) File
GPDI <- read_csv("~/ECO 530/Data/GPDI.csv")

```

```

Data2 <- data.frame(
  DATE = Data$DATE, # Assuming all lists have the same years
  Net_Trade = Net_Trade$`Net Trade`,
  GDP = GDP$GDP,
  GDPDEF = GDPDEF$GDPDEF,
  Interest_Rate = Interest_Rate$'FEDFUNDS',
  Gov_Expen = Gov_Expen$'Government Expenditure',
  UNRate = UNRate$"UNRATE",
  GPDI = GPDI$GPDI
)
# Sort chronologically
Data2 <- Data2 %>% arrange(DATE)

```

```

#Creating the lagged variables
Data2 <- Data2 %>%
  mutate( Lagged_GDP = dplyr::lag(GDP, 1),
          Lagged_Net_Trade = dplyr::lag(Net_Trade, 1) )

```

```

# Specify the file path and name
file_path <- "Data2.csv" # Create the CSV file
write.csv(Data2, file_path, row.names = FALSE)

```

```

Poster_Data <- read_csv("~/ECO 530/Data/Data2.csv")
# Convert the 'DATE' column to a proper date format
Poster_Data$DATE <- as.Date(Poster_Data$DATE, format = "%m/%d/%Y")

# Create a 'Years' column by extracting the year from the 'DATE'
Poster_Data$Years <- as.numeric(format(Poster_Data$DATE, "%Y"))

```

```
#Performing the ADF test on GDP and Nettrade  
adf.test(Poster_Data$GDP)
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: Poster_Data$GDP  
## Dickey-Fuller = 1.0032, Lag order = 4, p-value = 0.99  
## alternative hypothesis: stationary
```

```
adf.test(Poster_Data$Net_Trade)
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: Poster_Data$Net_Trade  
## Dickey-Fuller = -2.1191, Lag order = 4, p-value = 0.5271  
## alternative hypothesis: stationary
```

```
# List of control variables to check  
control_vars <- c("GDP", "GDPDEF", "Interest_Rate", "Gov_Expen", "UNRate", "GPDI")  
# Function to apply ADF test and print results  
check_stationarity <- function(var_name, data) { test_result <- adf.test(data[[var_name]], alter  
native = "stationary")  
print(paste("ADF Test for", var_name, ": p-value =", test_result$p.value)) }  
# Check stationarity for each control variable  
lapply(control_vars, check_stationarity, data = Poster_Data)
```

```
## [1] "ADF Test for GDP : p-value = 0.99"  
## [1] "ADF Test for GDPDEF : p-value = 0.788103802392348"  
## [1] "ADF Test for Interest_Rate : p-value = 0.416291697925408"  
## [1] "ADF Test for Gov_Expen : p-value = 0.454435931725984"  
## [1] "ADF Test for UNRate : p-value = 0.387514795626356"  
## [1] "ADF Test for GPDI : p-value = 0.859897192920853"
```

```
## [[1]]
## [1] "ADF Test for GDP : p-value = 0.99"
##
## [[2]]
## [1] "ADF Test for GDPDEF : p-value = 0.788103802392348"
##
## [[3]]
## [1] "ADF Test for Interest_Rate : p-value = 0.416291697925408"
##
## [[4]]
## [1] "ADF Test for Gov_Expen : p-value = 0.454435931725984"
##
## [[5]]
## [1] "ADF Test for UNRate : p-value = 0.387514795626356"
##
## [[6]]
## [1] "ADF Test for GPDI : p-value = 0.859897192920853"
```

```
reg1 <- lm_robust (Lagged_GDP ~ Lagged_Net_Trade, Poster_Data)
```

```
modelsummary(list("Model 1" =reg1) , gof_map = c("nobs"), stars = TRUE,
  fmt = 2, statistic= 'std.error',
  notes ="Robust SEs in parentheses",
  title = "Regression Analysis of Net Trade on GDP",
  output="kableExtra") %>% kable_classic_2()
```

Regression Analysis of Net Trade on GDP

	Model 1
(Intercept)	6423.41***
	(404.47)
Lagged_Net_Trade	-0.21***
	(0.01)
Num.Obs.	123
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
Robust SEs in parentheses	

```
reg2 <- lm_robust (Lagged_GDP ~ Lagged_Net_Trade + Years, Poster_Data)

modelsummary(list("Model 2" = reg2),
  gof_map = c("nobs"),
  stars = TRUE,
  statistic = 'std.error',
  notes = "Robust SEs in parentheses",
  title = "Regression Analysis of Net Trade on GDP",
  output = "kableExtra") %>%
kable_classic_2()
```

Regression Analysis of Net Trade on GDP

	 Model 2
(Intercept)	-19758.291
	(74590.243)
Lagged_Net_Trade	-0.208***
	(0.013)
Years	13.077
	(37.173)
Num.Obs.	123
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
Robust SEs in parentheses	

```
reg3 <- lm_robust(Lagged_GDP ~ Lagged_Net_Trade + Years + GDPDEF + Interest_Rate + Gov_Expen + U
NRate + GPDI, data = Poster_Data)

modelsummary(list("Model 3"=reg3) , gof_map = c("nobs"), stars = TRUE,
  statistic= 'std.error',
  notes ="Robust SEs in parentheses",
  output="kableExtra") %>% kable_classic()
```

	 Model 3
(Intercept)	6066.712
	(4794.134)
Lagged_Net_Trade	0.002
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
Robust SEs in parentheses	

	Model 3
	(0.003)
Years	-8.989***
	(2.319)
GDPDEF	293.138***
	(37.495)
Interest_Rate	-10.952
	(21.725)
Gov_Expen	0.268+
	(0.150)
UNRate	-185.783+
	(104.064)
GPGDI	0.336
	(0.565)
Num.Obs.	123

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Robust SEs in parentheses

```
calculate_vif <- function(reg3) {
  formula <- formula(reg3)
  variables <- all.vars(formula)[-1] # Exclude the dependent variable
  vif_values <- sapply(variables, function(var) {
    other_vars <- variables[variables != var]
    sub_model <- lm(as.formula(paste(var, "~", paste(other_vars, collapse = "+"))), data = Poster_Data)
    1 / (1 - summary(sub_model)$r.squared)
  })
  return(vif_values)
}

# Apply custom VIF calculation to lm_robust
vif_values <- calculate_vif(reg3)
print(vif_values)
```

```
## Lagged_Net_Trade      Years      GDPDEF      Interest_Rate
##      3.110310      1.555452      73.394465      3.062716
##      Gov_Expen      UNRate      GPDI
##      32.860862      14.397064      145.582025
```

```
dwtest(reg3)
```

```
##
## Durbin-Watson test
##
## data: reg3
## DW = 1.7125, p-value = 0.01455
## alternative hypothesis: true autocorrelation is greater than 0
```

```
cor(Poster_Data[, c("Lagged_GDP", "Lagged_Net_Trade", "Years", "GDPDEF", "Interest_Rate", "Gov_Expen", "UNRate", "GPDI")])
```

```
##      Lagged_GDP Lagged_Net_Trade      Years      GDPDEF
## Lagged_GDP      1      NA      NA      NA
## Lagged_Net_Trade      NA      1      NA      NA
## Years      NA      NA 1.0000000 0.2906845
## GDPDEF      NA      NA 0.2906845 1.0000000
## Interest_Rate      NA      NA 0.2125144 -0.5089509
## Gov_Expen      NA      NA 0.2504739 0.9688239
## UNRate      NA      NA -0.3562069 -0.0989169
## GPDI      NA      NA 0.3869680 0.9619948
##      Interest_Rate      Gov_Expen      UNRate      GPDI
## Lagged_GDP      NA      NA      NA      NA
## Lagged_Net_Trade      NA      NA      NA      NA
## Years      0.2125144 0.250473862 -0.356206899 0.3869680
## GDPDEF      -0.5089509 0.968823945 -0.098916900 0.9619948
## Interest_Rate      1.0000000 -0.525860347 -0.529016254 -0.3479537
## Gov_Expen      -0.5258603 1.000000000 0.006440544 0.9201810
## UNRate      -0.5290163 0.006440544 1.000000000 -0.3417681
## GPDI      -0.3479537 0.920181030 -0.341768138 1.0000000
```

```

# Ensure Poster_Data is sorted by 'Years' to maintain order
Poster_Data <- Poster_Data %>%
  arrange(Years)

# Create differenced variables, filling the first value with NA
Poster_Data <- Poster_Data %>%
  mutate(
    GDP_diff = c(NA, diff(GDP)),
    NetTrade_diff = c(NA, diff(Net_Trade)),
    GDPDEF_diff = c(NA, diff(GDPDEF)),
    Interest_Rate_diff = c(NA, diff(Interest_Rate)),
    Gov_Expen_diff = c(NA, diff(Gov_Expen)),
    UNRate_diff = c(NA, diff(UNRate)),
    GPDI_diff = c(NA, diff(GPDI)),
  )

# Remove the first row with NA values due to differencing
Poster_Data <- na.omit(Poster_Data)

# Check the structure of the filtered dataset to confirm changes
str(Poster_Data)

```

```

## tibble [123 × 18] (S3: tbl_df/tbl/data.frame)
##  $ DATE          : Date[1:123], format: "1993-10-01" "1993-04-01" ...
##  $ Net_Trade      : num [1:123] -33420 -60292 -41719 -6929 -32812 ...
##  $ GDP            : num [1:123] 10436 14899 18712 6809 10470 ...
##  $ GDPDEF         : num [1:123] 73.3 88.4 98.2 64 73.8 ...
##  $ Interest_Rate  : num [1:123] 6.4 1.81 0.38 3.04 5.31 0.16 0.4 3.09 3.97 0.18 ...
##  $ Gov_Expen      : num [1:123] 3210 5236 6318 2398 3281 ...
##  $ UNRate         : num [1:123] 3.9 6.1 4.9 7 4.3 7.3 5 6.7 4.5 8.7 ...
##  $ GPDI           : num [1:123] 2067 2498 3246 1094 1971 ...
##  $ Lagged_GDP     : num [1:123] 10318 14866 18526 6729 10436 ...
##  $ Lagged_Net_Trade : num [1:123] -33333 -60889 -36801 -6777 -33420 ...
##  $ Years          : num [1:123] 1993 1993 1993 1994 1994 ...
##  $ GDP_diff       : num [1:123] 3706 4463 3813 -11903 3661 ...
##  $ NetTrade_diff  : num [1:123] -26643 -26872 18573 34790 -25883 ...
##  $ GDPDEF_diff    : num [1:123] 9.71 15.06 9.76 -34.15 9.81 ...
##  $ Interest_Rate_diff: num [1:123] 3.33 -4.59 -1.43 2.66 2.27 -5.15 0.24 2.69 0.88 -3.79 ...
##  $ Gov_Expen_diff : num [1:123] 826 2026 1081 -3919 883 ...
##  $ UNRate_diff    : num [1:123] -3.1 2.2 -1.2 2.1 -2.7 3 -2.3 1.7 -2.2 4.2 ...
##  $ GPDI_diff      : num [1:123] 983 431 748 -2152 877 ...
##  - attr(*, "na.action")= 'omit' Named int 1
##  ... attr(*, "names")= chr "1"

```

```

# List of differenced variables to check
differenced_vars <- c("GDP_diff", "GDPDEF_diff", "Interest_Rate_diff", "Gov_Expen_diff", "UNRate_diff", "GPDI_diff")

# Check stationarity for each differenced variable
lapply(differenced_vars, check_stationarity, data = Poster_Data)

```

```
## [1] "ADF Test for GDP_diff : p-value = 0.01"
## [1] "ADF Test for GDPDEF_diff : p-value = 0.01"
## [1] "ADF Test for Interest_Rate_diff : p-value = 0.01"
## [1] "ADF Test for Gov_Expen_diff : p-value = 0.01"
## [1] "ADF Test for UNRate_diff : p-value = 0.01"
## [1] "ADF Test for GPDI_diff : p-value = 0.01"
```

```
## [[1]]
## [1] "ADF Test for GDP_diff : p-value = 0.01"
##
## [[2]]
## [1] "ADF Test for GDPDEF_diff : p-value = 0.01"
##
## [[3]]
## [1] "ADF Test for Interest_Rate_diff : p-value = 0.01"
##
## [[4]]
## [1] "ADF Test for Gov_Expen_diff : p-value = 0.01"
##
## [[5]]
## [1] "ADF Test for UNRate_diff : p-value = 0.01"
##
## [[6]]
## [1] "ADF Test for GPDI_diff : p-value = 0.01"
```

```
# Run the regression model with the differenced variables
reg4 <- lm_robust(
  GDP_diff ~ NetTrade_diff + Years + GDPDEF_diff + Interest_Rate_diff + Gov_Expen_diff + UNRate_diff + GPDI_diff,
  data = Poster_Data
)

# Summarize the regression results
modelsummary(list("Model 4"=reg4) , gof_map = c("nobs"), stars = TRUE,
  statistic= 'std.error',
  notes ="Robust SEs in parentheses",
  output="kableExtra") %>% kable_classic()
```

 Model 4	
(Intercept)	-1849.104
	(4381.898)
NetTrade_diff	0.001
	(0.001)
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
Robust SEs in parentheses	

	Model 4
Years	0.921
	(2.182)
GDPDEF_diff	250.937***
	(11.193)
Interest_Rate_diff	-23.416*
	(10.491)
Gov_Expen_diff	0.217***
	(0.058)
UNRate_diff	-101.050*
	(41.361)
GPDI_diff	1.203***
	(0.210)
Num.Obs.	123

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Robust SEs in parentheses

```
calculate_vif <- function(reg4) {
  formula <- formula(reg4)
  variables <- all.vars(formula)[-1] # Exclude the dependent variable
  vif_values <- sapply(variables, function(var) {
    other_vars <- variables[variables != var]
    sub_model <- lm(as.formula(paste(var, "~", paste(other_vars, collapse = "+"))), data = Poster_Data)
    1 / (1 - summary(sub_model)$r.squared)
  })
  return(vif_values)
}

# Apply custom VIF calculation to lm_robust
vif_values2 <- calculate_vif(reg4)
print(vif_values2)
```

##	NetTrade_diff	Years	GDPDEF_diff	Interest_Rate_diff
##	3.706278	1.001863	87.325083	2.387236
##	Gov_Expen_diff	UNRate_diff	GPDI_diff	
##	33.256887	17.903330	186.441570	